

Yujian Cao

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Education

University of Liverpool , PhD in Computer Science <i>Supervised by Dr. Shufang Zhu and Prof. Sven Schewe</i>	Sep 2025 – Present
Hong Kong Baptist University , MA in Artificial Intelligence and Digital Media	Sep 2022 – Jul 2024
Anhui Agricultural University , BEng in Computer Science and Technology	Sep 2018 – Jun 2022
Anhui Normal University , BA in Chinese Language and Literature (<i>Part-time</i>)	Aug 2019 – Dec 2022

Research Interest

My current research focuses on formal methods (FM), particularly reactive synthesis with linear temporal logic on finite traces (LTLf). In the future, I plan to explore the interdisciplinary research area of artificial intelligence (AI) and formal methods, focusing on the development of trustworthy autonomous AI systems.

Experience

IT Technical Support , FESTO – Shanghai, China	Mar 2025 – Jul 2025
Network Engineer , China Unicom – Anhui, China	Jul 2022 – Jan 2025

Publications

Optimal LTLf Synthesis <i>Yujian Cao, Sven Schewe, Qiyi Tang, Shufang Zhu</i> <i>Accepted at IJCAI 2026</i>	2026
Coverage Games with LTLf Objectives <i>Yujian Cao, Sven Schewe, Qiyi Tang, Shufang Zhu</i> <i>Under Review at CONCUR 2026</i>	2026

Skills

Languages: Chinese (Native), English (IELTS 6.5), Korean (TOPIK 1), Japanese (Basic)

Certifications:

- Senior High School Teacher Qualification Certificate (MOE, China)
- Network Engineer Certificate (MOHRSS, China)
- Junior Accountant Certificate (MOF, China)